

Total marks (36)**Attempt questions 1 – 3****All questions are of equal value****Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.****Question 1 (12 marks)** Use a SEPARATE writing booklet**Marks**(a) Consider the expansion of $\left(3x^2 - \frac{1}{x}\right)^9$.

(i) How many terms are there in this expansion?

1

(ii) Find the constant term in this expansion.

3(b) By equating the coefficients of x^5 on both sides of the identity

$$(1+x)^4(1+x)^4 = (1+x)^8$$

prove that ${}^4C_0 \times {}^4C_1 + {}^4C_1 \times {}^4C_2 + {}^4C_2 \times {}^4C_3 + {}^4C_3 \times {}^4C_4 = \frac{8!}{3! \times 5!}$

4(c) Using the expansion of $(1+x)^n$ prove:

$$(i) \quad 10^n = \binom{n}{0} + 3^2 \cdot \binom{n}{1} + 3^4 \cdot \binom{n}{2} + \dots + 3^{2n} \cdot \binom{n}{n}$$

1

$$(ii) \quad \text{Hence, show that } 1 + 3^4 \cdot \binom{n}{2} + 3^8 \cdot \binom{n}{4} + \dots + 3^{2n} \cdot \binom{n}{n} = 2^{n-1}(5^n + 4^n),$$

where n is an even integer.**3**

Question 2 (12 marks)

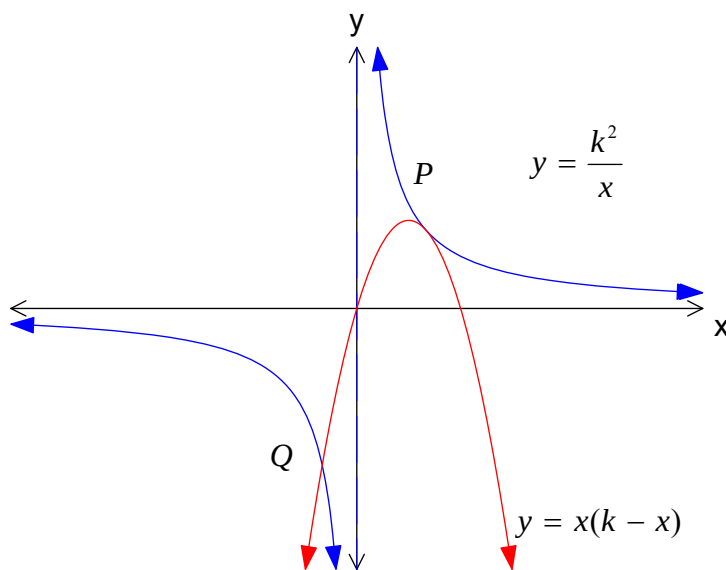
Use a SEPARATE writing booklet

Marks

- (a) (i) Find the linear factors of $6 + 5x - 2x^2 - x^3$. **2**
- (ii) Find the values of x for which $6 + 5x - 2x^2 - x^3 > 0$. **2**

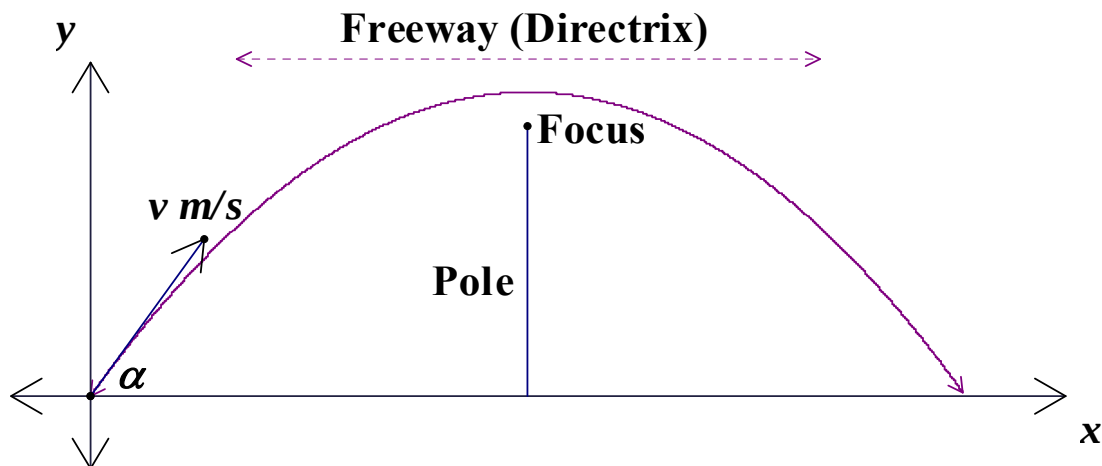
- (b) The polynomial $P(x) = 2x^3 + ax^2 + bx + 6$ has $(x - 1)$ as a factor and leaves a remainder of -12 when divided by $(x + 2)$. Find the values of a and b . **3**

(c)



In the diagram above, the curves $y = \frac{k^2}{x}$ and $y = x(k - x)$, where $k > 0$, touch at the point P and intersect at the point Q .

- (i) Explain why the equation $x^3 - kx^2 + k^2 = 0$ has real roots α , α , and β for some $\alpha \neq \beta$? **2**
- (ii) Find the exact values of k , α and β . **3**



A frog jumps with initial velocity v m/s at an angle of projection α and its path traces a parabolic arc as shown above. The frog's horizontal displacement from the origin, t seconds after jumping is given by the equation $x = vt \cos \alpha$ [Do not prove this]. The frog's vertical displacement from the origin, t seconds after jumping is given by the equation $y = vt \sin \alpha - \frac{1}{2}gt^2$ [Do not prove this].

- (i) Show that the frog lands after $\frac{2v \sin \alpha}{g}$ seconds. 2
- (ii) Show that the frog's range is $\frac{v^2 \sin 2\alpha}{g}$ metres. 2
- (iii) Show that the frog's maximum height is $\frac{v^2 \sin^2 \alpha}{2g}$ metres. 2
- (iv) Show that the ratio of the frog's maximum height to range is $\frac{\tan \alpha}{4}$. 2
- (v) Let $\alpha_{\max}$ be the angle the frog jumps at to ensure maximum range. Find the ratio of the frog's maximum height to its range in this case. 1
- (vi) Let $g = 10 \text{ m/s}^2$ and $v = \sqrt{85} \text{ m/s}$.
 Let α_{equal} be the angle where the frog's maximum height equals its range.
 At this angle, the frog needs a 40cm gap to squeeze between a pole and a freeway.
 The top of the pole is the focus of the parabola and the freeway is the directrix.
 Can the frog squeeze through? 3

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note $\ln x = \log_e x, \quad x > 0$

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Q1 (a) (i) 10 terms

$$\begin{aligned} \text{(ii)} \quad \left(3x^2 - \frac{1}{x}\right)^9 &= \sum_{k=0}^9 (3x^2)^k \left(\frac{-1}{x}\right)^{9-k} {}^9C_k \\ &= \sum_{k=0}^9 {}^9C_k \cdot 3^k x^{2k} (-1)^{9-k} x^{k-9} \\ &= \sum_{k=0}^9 {}^9C_k (-1)^{9-k} 3^k x^{3k-9} \end{aligned}$$

for the constant term : $3k-9=0$

$$\therefore k=3$$

$$\begin{aligned} \therefore \text{coefficient} &= (-1)^6 {}^9C_3 \cdot 3^3 \\ &= 2268 \end{aligned}$$

$$\text{(b) LHS} = \left({}^4C_0 + {}^4C_1 x + {}^4C_2 x^2 + {}^4C_3 x^3 + {}^4C_4 x^4\right)^2$$

$$\therefore \text{co-eff. of } x^5 = {}^4C_1 x {}^4C_4 + {}^4C_2 x {}^4C_3 + {}^4C_3 x {}^4C_2 + {}^4C_4 x {}^4C_1$$

$$= {}^4C_1 x {}^4C_0 + {}^4C_2 x {}^4C_1 + {}^4C_2 x {}^4C_3 + {}^4C_3 x {}^4C_4$$

(since ${}^nC_r = {}^nC_{n-r}$)

$$\text{RHS} = \sum_{k=0}^8 x^k \cdot {}^8C_k$$

$$\therefore \text{co-eff. of } x^5 = {}^8C_5$$

$$= \frac{8!}{(8-5)! 5!}$$

$$= \frac{8!}{3! 5!}$$

$$\therefore {}^4C_0 x {}^4C_1 + {}^4C_1 x {}^4C_2 + {}^4C_2 x {}^4C_3 + {}^4C_3 x {}^4C_4 = \frac{8!}{3! 5!}$$

$$(c)(i) \quad (1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$$

$$\text{let } x = 9 = 3^2$$

$$\therefore 10^n = \binom{n}{0} + \binom{n}{1}3^2 + \binom{n}{2}3^4 + \dots + \binom{n}{n}3^{2n} \quad (1)$$

$$(ii) \text{ let } x = -9 = -3^2$$

$$\therefore (-8)^n = \binom{n}{0} - \binom{n}{1}3^2 + \binom{n}{2}3^4 - \dots + \binom{n}{n}3^{2n} \quad (2)$$

but since n is even then $(-8)^n = 8^n$

adding (1) and (2) :

$$2\binom{n}{0} + 2\binom{n}{2}3^4 + \dots + 2\binom{n}{n}3^{2n} = 10^n + 8^n$$

$$\therefore \binom{n}{0} + \binom{n}{2}3^4 + \dots + \binom{n}{n}3^{2n} = \frac{10^n + 8^n}{2}$$

$$= \frac{2^n \cdot 5^n + 2^n \cdot 4^n}{2}$$

$$= 2^{n-1} \cdot 5^n + 2^{n-1} \cdot 4^n$$

$$= 2^{n-1}(5^n + 4^n)$$

Solutions

Q2

(a) (i) $P(x) = 6 + 5x - 2x^2 - x^3$

$$P(-1) = 6 - 5 - 2 + 1 = 0$$

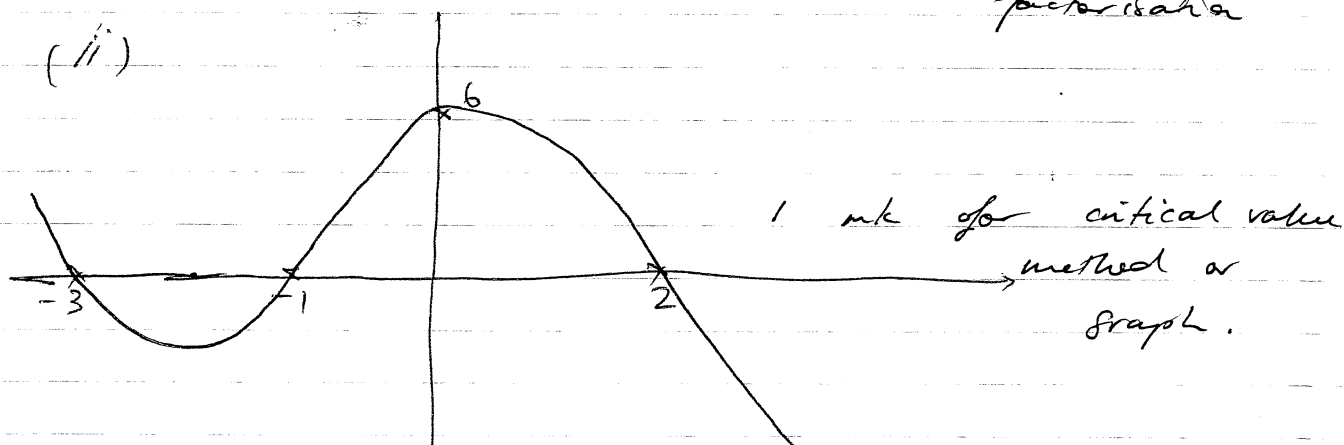
$\therefore (x+1)$ is a factor $\therefore$ 1 mark for factor then.

$$\begin{array}{r} -x^2 - x + 6 \\ x+1 \overline{) -x^3 - 2x^2 + 5x + 6} \\ \underline{-x^3 - x^2} \\ -x^2 + 5x \\ \underline{-x^2 - x} \\ 6x + 6 \\ \underline{6x + 6} \\ 0 \end{array} \Rightarrow 6 - x - x^2 = (x+3)(2-x).$$

$$\therefore P(x) = (3+x)(2-x)(x+1) = (x+1)(x+3)(2-x).$$

1 mark for continued factorisation

(ii)



$\therefore P(x) > 0$ if $x < -3$ or $-1 < x < 2$.

1 mark for marks

(b) $P(x) = 2x^3 + ax^2 + bx + 6$

$$P(1) = 2 + a + b + 6$$

$$= a + b + 8.$$

$$P(-2) = -16 + 4a - 2b + 6$$

$$= 4a - 2b - 10$$

$$\therefore 4a - 2b - 10 = -12$$

$$4a - 2b = -2$$

$$2a - b = -1 \quad \dots (1)$$

$$2a - b = -1 \quad \dots (2)$$

$$\therefore 3a = -9$$

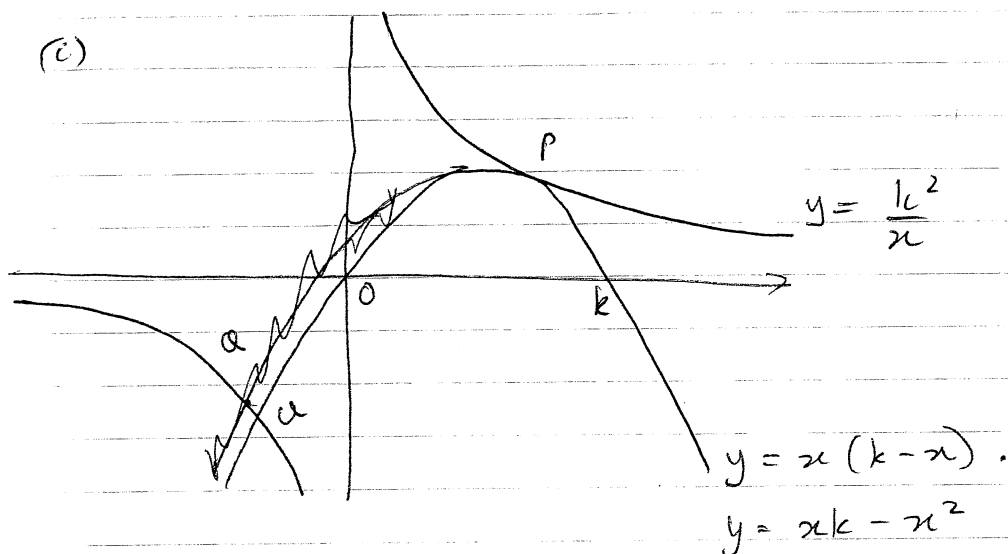
$$a = -3$$

$$b = -5$$

$$b = -5$$

1 mark
both correct.

(c)



(i) If $xk - x^2 = \frac{k^2}{x}$

then $x^2k - x^3 = k^2$
 $x^3 - kx^2 + k^2 = 0$.

ie. the intersection of the two curves.

double root at $x = P$ and linear root at Q
 hence α, α, β are the roots.

(ii) $\alpha^2 + \beta = k^2$

and

and

and

$$\alpha^2\beta = -k^2 \quad \dots (1)$$

$$2\alpha + \beta = k \quad \dots (2)$$

$$\alpha^2 + 2\alpha\beta = 0$$

$$\alpha(\alpha + 2\beta) = 0 \quad \dots (3)$$

if $\alpha = 0$ then β is the only solution so
 $\alpha \neq 0$

$$\therefore \alpha = -2\beta$$

$$k = k + 4\beta$$

$$\therefore -3\beta = k$$

$$\beta = -\frac{k}{3}$$

$$\therefore \alpha = \frac{2k}{3}$$

Now

$$\alpha^2\beta = k^2$$

$$\therefore \frac{4k^2}{9} \cdot \left(-\frac{k}{3}\right) = k^2$$

$$\therefore -\frac{4k^3}{27} = k^2$$

$$\therefore k = -\frac{27}{4}$$

$$\therefore \alpha = -\frac{9}{2}$$

$$\beta = \frac{9}{4}$$

$$\frac{4k^2}{9} \cdot \frac{k}{3} = k^2$$

$$\therefore 4k^3 - 12k^2 = 0$$

$$\therefore k^2(4k - 12) = 0$$

$$k = \frac{12}{4} = 3$$

$$\therefore \alpha = -\frac{27}{4}, \beta = \frac{9}{4}$$

QUESTION THREE

(i) Frog lands when $y=0$

$$\therefore v t \sin \alpha - \frac{1}{2} g t^2 = 0 \quad \checkmark$$

$$\therefore t (v \sin \alpha - \frac{1}{2} g t) = 0$$

$$\therefore t=0 \text{ or } t = \frac{2v \sin \alpha}{g} \quad \checkmark$$

$$\therefore \text{Frog lands when } t = \frac{2v \sin \alpha}{g}$$

(ii) When $t = \frac{2v \sin \alpha}{g}$

$$x = v t \cos \alpha$$

$$= v \cos \alpha \left(\frac{2v \sin \alpha}{g} \right) \quad \checkmark$$

$$= \frac{v^2}{g} (2 \sin \alpha \cos \alpha)$$

$$= \frac{v^2 \sin 2\alpha}{g} \quad \checkmark$$

(iii) maximum height occurs when $\dot{y}=0$

$$\dot{y} = v \sin \alpha - g t$$

$$= 0 \text{ when } t = \frac{v \sin \alpha}{g} \quad \checkmark$$

$$y = v t \sin \alpha - \frac{1}{2} g t^2$$

$$= \frac{v^2 \sin^2 \alpha}{g} - \frac{v^2 \sin^2 \alpha}{2g}$$

$$= \frac{v^2 \sin^2 \alpha}{2g} \quad \checkmark$$

(iv) $\frac{\text{height}}{\text{range}} = \frac{\frac{v^2 \sin^2 \alpha}{2g}}{\frac{v^2 \sin 2\alpha}{g}} \quad \checkmark$

$$= \frac{\sin^2 \alpha}{4 \sin \alpha \cos \alpha}$$

$$= \frac{\tan \alpha}{4} \quad \checkmark$$

(v) maximum range occurs when $\sin 2\alpha = 1$

$$\text{i.e. when } \alpha = \frac{\pi}{4}$$

$$\therefore \frac{\text{height}}{\text{range}} = \frac{1}{4} \quad \checkmark$$

(vi) if maximum height = range then $\tan \alpha = 4$

$$x = v t \cos \alpha \quad \therefore t = \frac{x}{v \cos \alpha}$$

$$y = v t \sin \alpha - \frac{1}{2} g t^2$$

$$= v \sin \alpha \left(\frac{x}{v \cos \alpha} \right) - \frac{1}{2} g \left(\frac{x}{v \cos \alpha} \right)^2$$

$$= x \tan \alpha - \frac{1}{2} \frac{g x^2}{v^2 \sec^2 \alpha}$$

$$\therefore y = 4x - \frac{5x^2}{85} \times 17$$

$$y = 4x - x^2 \quad \checkmark$$

$$x^2 - 4x + 4 = -y + 4$$

$$(x-2)^2 = -(y-4)$$

$$\therefore \text{Vertex } (2, 4) \text{ focal length } \frac{1}{4}$$

$$\text{focus } (2, 3\frac{3}{4}) \text{ directrix: } y = 4\frac{1}{4} \quad \checkmark$$

$\therefore$ distance between focal length and directrix is $\frac{1}{2}$ metre $>$ 40cm

$\therefore$ Frog can squeeze through $\checkmark$

N.B. Had to get to eqn. of parabola for first mark.

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